

SRESHTH TIWARI

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[LinkedIn](#) · [GitHub](#)

EDUCATION

Dartmouth College, Hanover, NH

September 2024 - June 2028

Bachelor's in Computer Science and Mathematics

GPA 3.9

Relevant Coursework: *Computational Aspects of Robotics; Machine Learning and Statistical Data Analysis; Software Design and Implementation; Object Oriented Programming; Linear Algebra with Applications*

Honors: **Rufus Choate Scholar** (Academic Performance), **Dean's Honor List** (First Distinction), **Math Department Citation of Merit**, **Computer Science Department Citation of Merit**

EXPERIENCE

Citadel Securities, Quant Trading Invitational Participant

April 2026

- Developed automated trading strategy in Python using Rotman Interactive Trader API to execute trades for simulated trading competition.
- Placed 3rd out of 78 participants in overall PnL after 4 rounds of electronic trading.

Song Lab at Dartmouth College, Developer Intern

Feb 2026 - Aug 2026

- Developing large-scale metagenomics classification software in C++.
- Implemented a Paired 512/64K bit vector data structure to replace existing storage methods.
- Achieved ~19% faster rank query times on large (GB Scale) indices compared to previous implementations.

Metanoia Kleos Foundation, Software Engineer Intern

Jun 2025 - Sep 2025

- Developed an internal project management tool using TypeScript, Next.js, NestJS, and PostgreSQL.
- Implemented RESTful APIs and relational database schemas to support scalable data access across the application.
- Configured cloud infrastructure and deployment workflows on AWS; reduced manual operational effort by 35% and improved release turnaround by 60%.

PROJECTS

LLM-Augmented Autonomous Robot Navigation

[Github](#)

- Used ROS2 to develop an autonomous navigation pipeline integrating A* path planning with an LLM-based reasoning layer to dynamically replan trajectories when unexpected obstacles invalidate the robot's original route.
- Built a simulation and validation framework for dynamic navigation scenarios, injecting obstacles during execution and evaluating baseline versus LLM-augmented planners on path efficiency, successful replanning, and navigation behavior.
- Integrated robot localization, occupancy-grid mapping, and collision-aware path generation into the planning pipeline, translating simulated environment state into actionable navigation decisions.

"@Log": Attendance Tracking and Transport Management App for Schools

[App Store](#)

- Developed a full-stack web app for school administration to manage student attendance and transport data using React, Node.js, and Express.js.
- Built the backend with AWS, SQL, and MongoDB. Currently deployed across 50+ schools with 100K+ users.

Computational Portrait Photography iOS App

[Github](#)

- Developed an iOS computational portrait photography application using React Native with native Swift modules, AVFoundation, Vision, and Core Image.
- Implemented native iOS camera capture and image processing modules for subject segmentation, background blur synthesis, blue detection, and sharpening based restoration.

LEADERSHIP & ACTIVITIES

Teaching Assistant, Computer Science 29.06 – Digital Tangible User Interfaces

Jun 2026 - Present

International Collegiate Programming Competition, Dartmouth College

Sep 2025 - Present

Dartmouth Quant Club

Sep 2024 - Present

SKILLS

Languages/Frameworks: Python, C++, JavaScript/TypeScript, Swift, SQL, React, React Native, Node.js, Express.js, Next.js

Tools/Technologies: PostgreSQL, MongoDB, Prisma, Git, Linux/Unix, AWS, AVFoundation, Vision, Core Image, ROS2,

PyTorch, Computer Vision, A* Path Planning, Localization, Occupancy-Grid Mapping, LLMs, Deep Learning